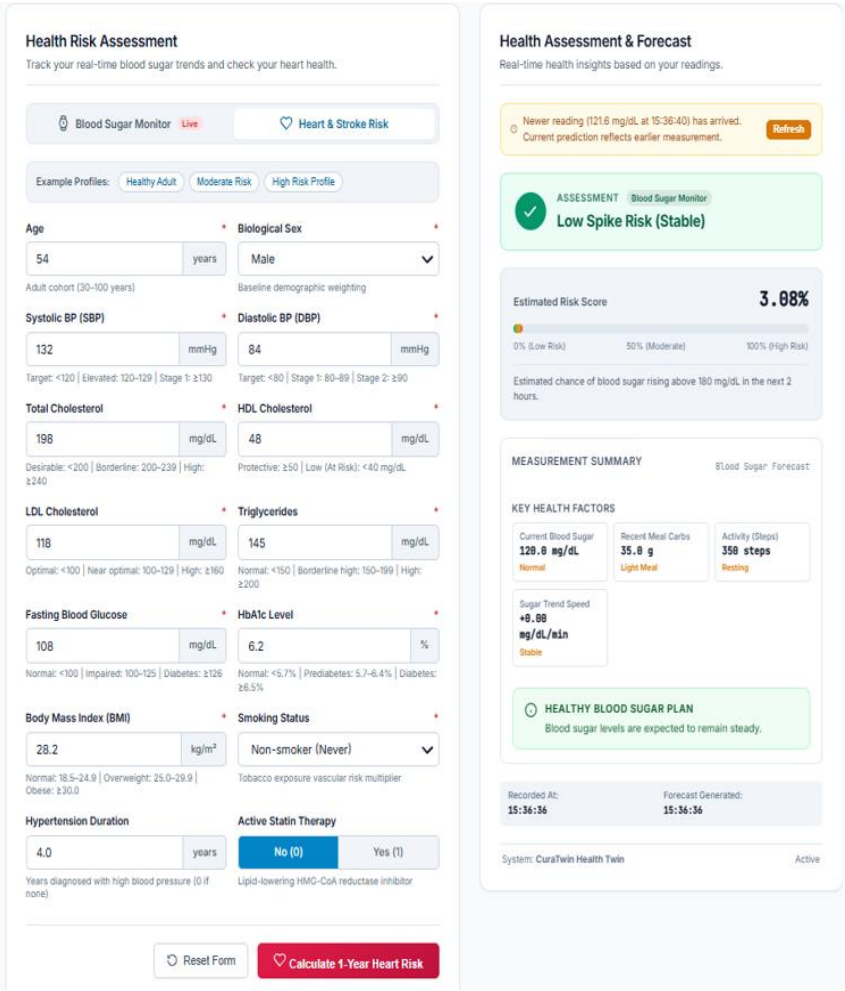
# CuraTwin: An AI-Powered Digital Twin for Type 2 Diabetes Monitoring




*Our Proposed Solution*

CuraTwin is an AI-powered healthcare Digital Twin designed to go beyond conventional blood glucose monitoring by integrating **Type 2 diabetes assessment, glucose trend monitoring, and cardiovascular risk prediction** into a unified platform.

The system combines diabetes-related parameters, historical health records, and physiological measurements to provide a broader understanding of a patient's health. Alongside monitoring blood glucose levels and indicators such as HbA1c, CuraTwin considers cardiovascular risk factors, including blood pressure, cholesterol levels, body mass index (BMI), age, and smoking status, to assess the potential risk of heart disease and stroke.

By bringing these interconnected health indicators together, CuraTwin aims to help users understand not only their current glucose status but also how diabetes and associated risk factors may affect their long-term cardiovascular health.

**Key Capabilities**

**1. Type 2 Diabetes Monitoring and Assessment**

- Monitor blood glucose levels and visualize glucose trends.
- Incorporate diabetes-related indicators, including fasting blood glucose and HbA1c.
- Present health readings alongside relevant lifestyle information, such as meals and physical activity, where available.

**2. Heart Disease and Stroke Risk Assessment**

- Assess cardiovascular risk using parameters such as systolic and diastolic blood pressure, total cholesterol, HDL, LDL, triglycerides, BMI, age, and smoking status.
- Incorporate relevant clinical factors, including hypertension history and statin therapy, where supported.
- Display an estimated risk score and assessment results based on the implemented model.

**3. Integrated Patient Digital Twin**

- Bring diabetes monitoring and cardiovascular risk assessment together in one interface.
- Combine available patient records with changing physiological measurements.
- Present health trends, risk assessments, and relevant indicators through an interactive dashboard.

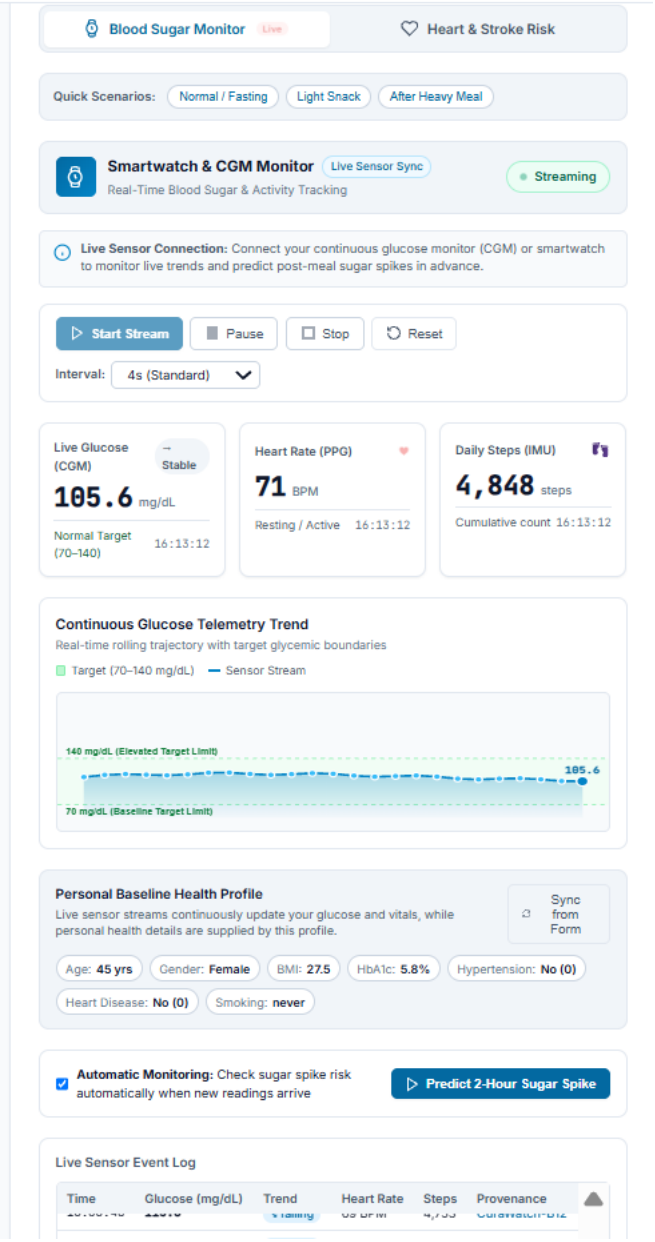
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*What Makes CuraTwin Different?*

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- **Dual Health Assessment:** Combines Type 2 diabetes monitoring with heart disease and stroke risk assessment.
- **Multiple Health Parameters:** Considers glucose, HbA1c, blood pressure, cholesterol, BMI, and lifestyle factors.
- **Integrated Digital Twin:** Brings historical health records and available physiological measurements into one platform.
- **Beyond Basic Monitoring:** Transforms health measurements into visual trends and estimated risk scores.
- **Proactive Healthcare Approach**: Aims to support earlier identification of potential complications and more informed health management.

## *Real-Time Sensor Monitoring*




CuraTwin currently uses simulated sensor data to demonstrate continuous glucose and physiological monitoring at configurable intervals, such as every 2 or 5 seconds. The system visualizes changing readings, tracks glucose trends, and supports the demonstration of post-meal glucose spike prediction.

Following the hackathon guidelines, the simulation can be extended using open-source wearable datasets or synthetic time-series data. Compatible physical sensors and wearable devices could also be integrated to collect real patient measurements, enabling a transition from simulated monitoring to real-time health tracking.

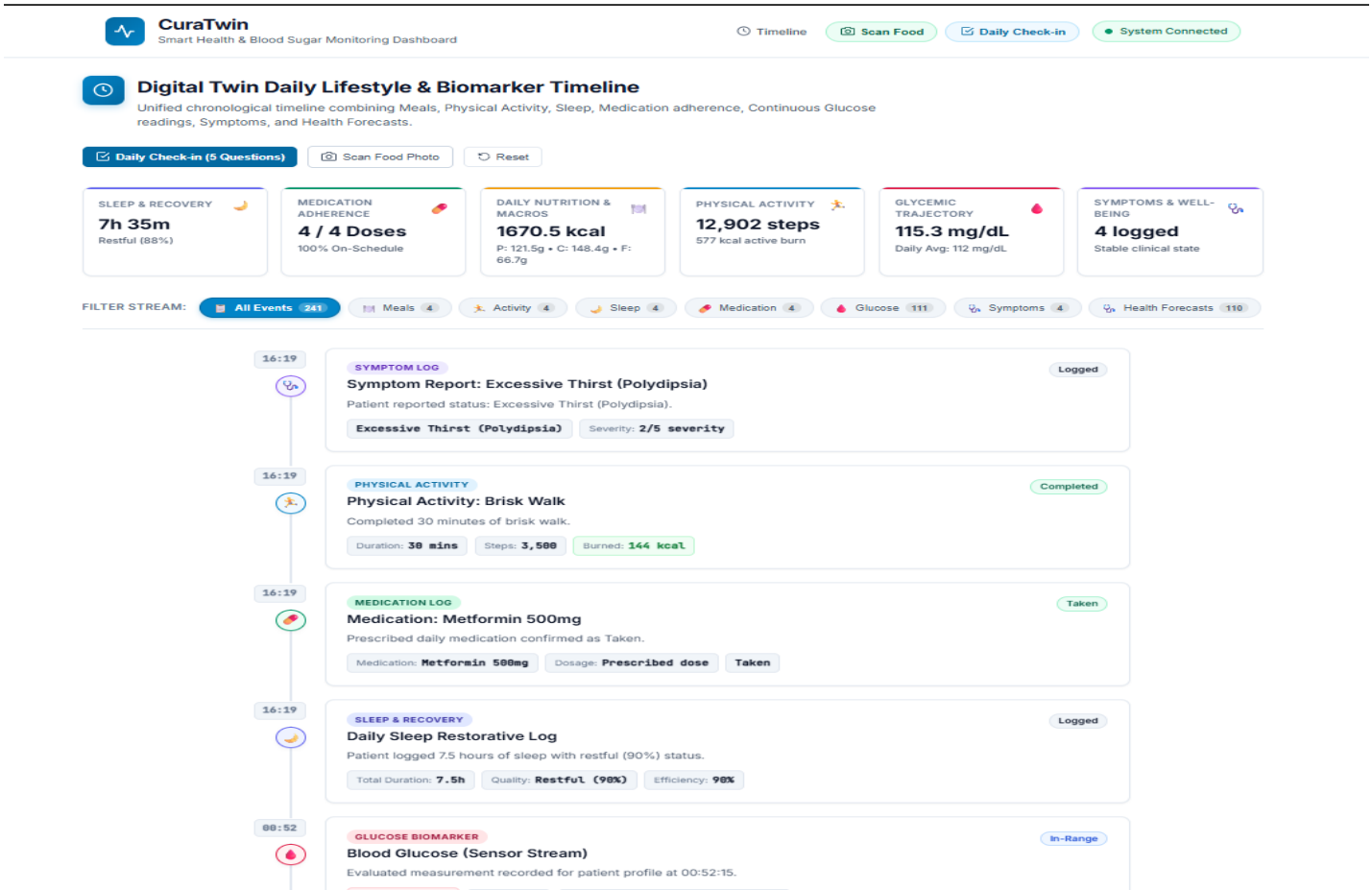


*Digital Twin: Daily Lifestyle & Biomarker Timeline*

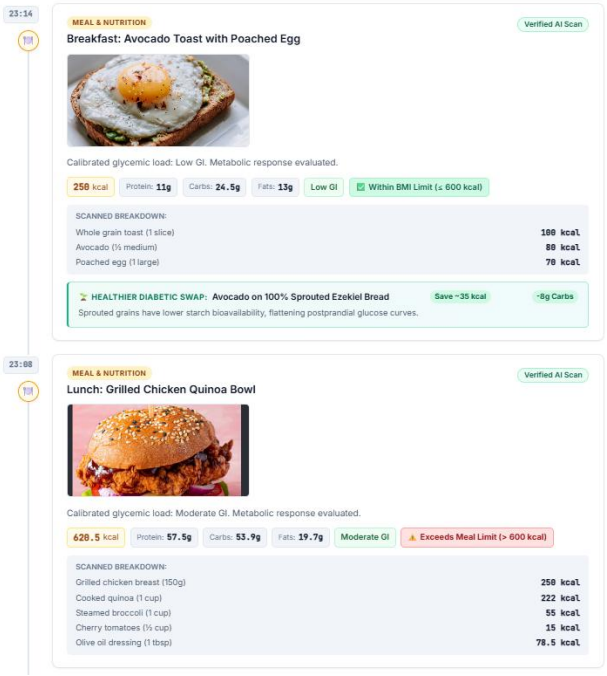
CuraTwin records patient inputs and organizes daily health information into a unified timeline, tracking **meals and nutrition, physical activity, sleep, medication adherence, glucose readings, symptoms, and health forecasts**. Based on these inputs and available sensor data, it updates the patient's health profile and generates forecasts to support personalized monitoring and risk assessment.

**Patient Input & Lifestyle Tracking**

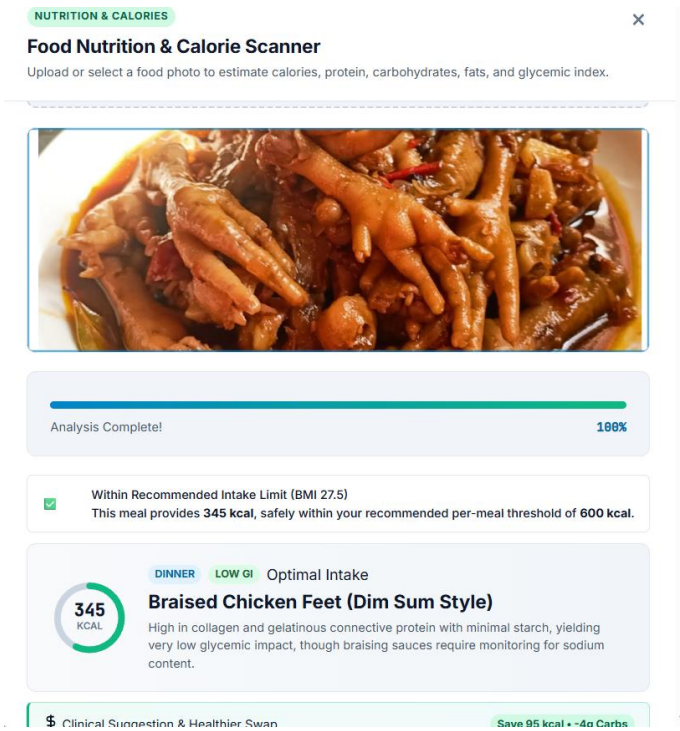
Patients can log their daily activities, record meals using the food-scanning feature, complete daily check-ins, and report symptoms or medication intake. These inputs are combined with glucose and physiological measurements to help CuraTwin analyze changing health patterns and provide relevant health forecasts.



## *Adding Meals*

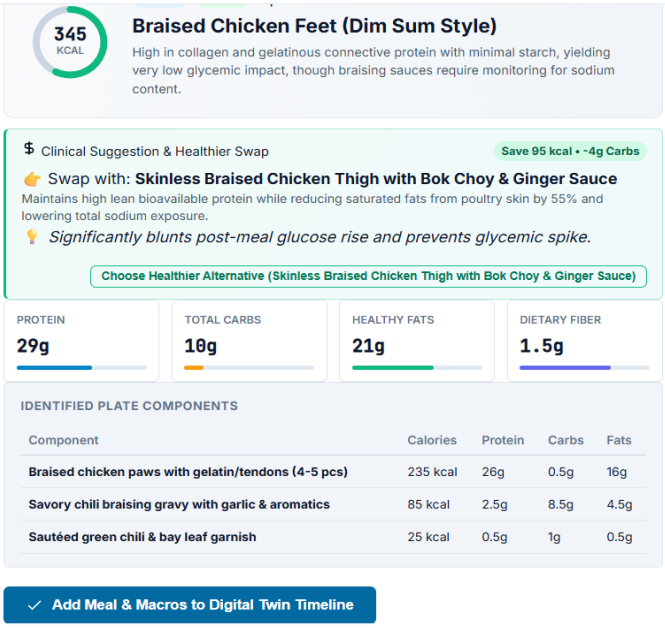



1. CuraTwin records meals in the patient's daily timeline, displaying estimated calories, protein, carbohydrates, fats, and glycemic index. Each meal is broken down into individual food components, helping patients understand their nutritional intake and its potential impact on blood glucose levels.




2. Patients can upload or scan a food image to receive an analysis of the identified dish, including estimated calories, macronutrients, and portion details. The system presents the results alongside dietary guidance relevant to diabetes management.

Based on the meal analysis, CuraTwin suggests healthier alternatives that may reduce carbohydrate intake or improve nutritional balance. It also displays estimated nutritional differences, helping patients make more informed food choices.



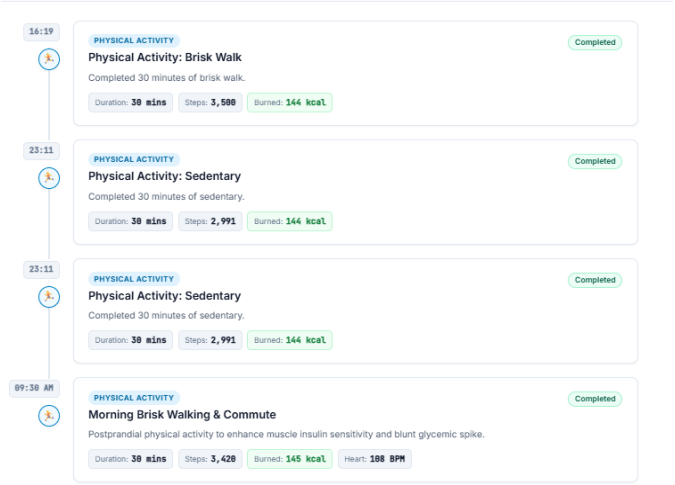

3. After analysis, patients can add the meal and its nutritional information to their Digital Twin timeline. This allows CuraTwin to maintain a record of dietary habits alongside glucose

readings, physical activity, sleep, and medication adherence for a more comprehensive view of daily health.

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*Daily Lifestyle & Medication Tracking*

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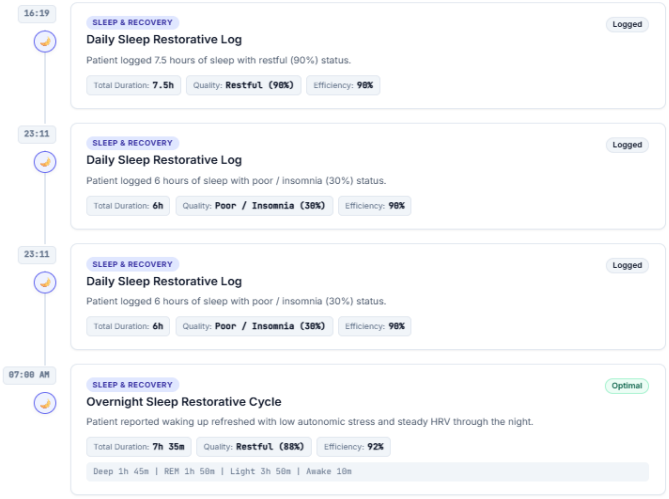


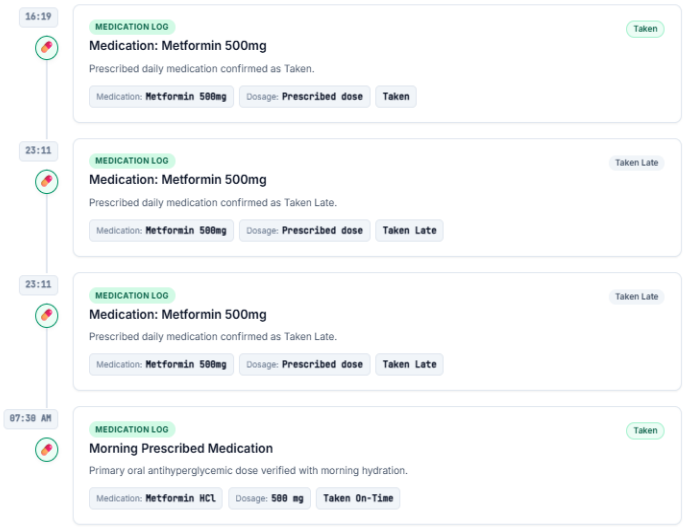

### 1. Physical Activity Tracking

CuraTwin records daily physical activities, including walking, exercise, and sedentary periods. It tracks duration, steps, and estimated calories burned, helping patients understand their activity patterns and their potential impact on glucose regulation.

### 2. Sleep and Recovery Monitoring

The system maintains sleep records, including sleep duration, quality, and recovery indicators. By tracking changes in sleep patterns, CuraTwin helps patients understand how rest may influence overall health and diabetes management.

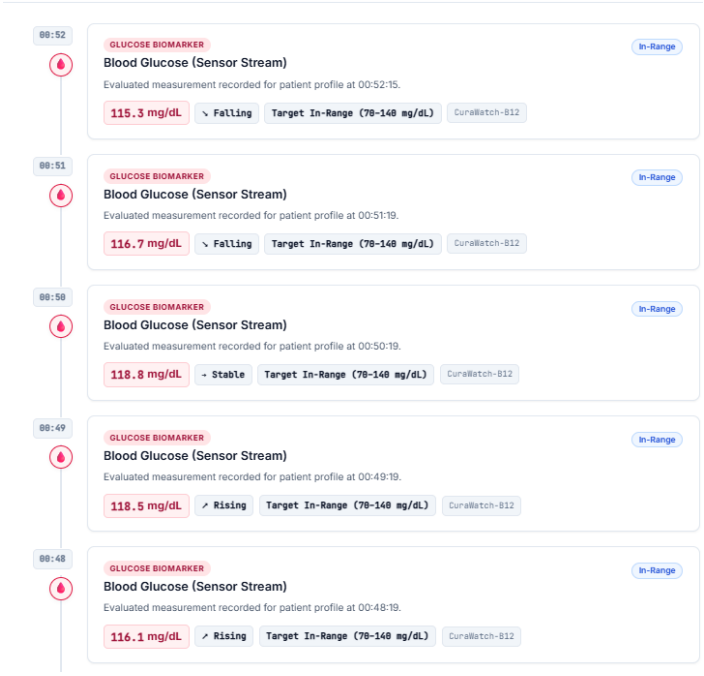



### 3. Medication Adherence Tracking

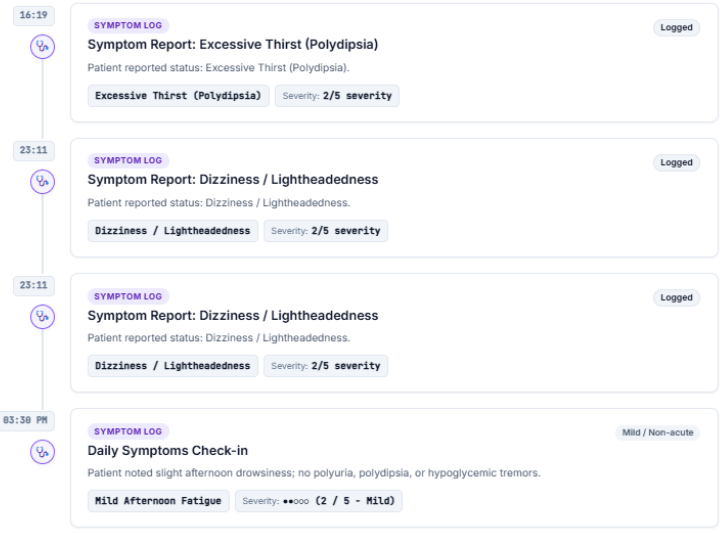
CuraTwin logs prescribed medications and records whether doses are taken on time, taken late, or missed, where supported. This helps patients monitor their medication routine and identify adherence patterns alongside their other health measurements.

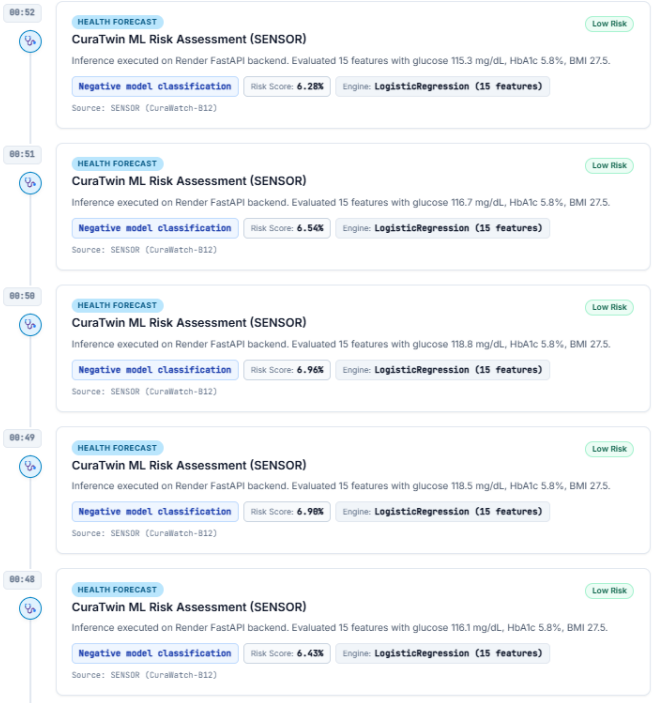
*Glucose Monitoring, Symptom Tracking & Health Forecasts*




**4. Continuous Glucose Monitoring:** Tracks changing glucose readings over time, displaying current levels, trends, timestamps, and target ranges using simulated sensor data.

**5.Symptom Tracking:** Records patient-reported symptoms such as excessive thirst, dizziness, and fatigue, including their severity, to provide additional context for health assessment.




6. **AI-Powered Health Forecasts:** Processes available health parameters through the implemented machine learning model and displays classification results and risk indicators alongside glucose readings.

## *How CuraTwin Can Help Patients*

- **Personalized Health Monitoring:** Tracks glucose levels, meals, physical activity, sleep, medication adherence, and symptoms in one place.
- **Early Awareness of Potential Risks:** Helps patients understand diabetes-related health patterns and cardiovascular risk factors associated with heart disease and stroke.
- **Better Daily Decisions:** Provides nutritional information, healthier food alternatives, and insights into lifestyle habits that may affect glucose control.
- **Continuous Health History:** Maintains a chronological record of health measurements and daily activities, making changes and recurring patterns easier to review.
- **Improved Patient–Doctor Communication:** Can help patients share their health timeline with doctors, giving clinicians a broader view of their daily habits, symptoms, medication routines, and recorded measurements.

## *Future Scope*

- **Doctor Dashboard:** Develop a clinician-facing interface where authorized doctors can review patient timelines, glucose trends, symptoms, medication adherence, and risk assessments remotely.
- **Real-Time Device Integration:** Connect compatible continuous glucose monitors, smartwatches, and blood pressure monitors to replace simulated readings with actual measurements.
- **Personalized Alerts:** Introduce configurable alerts for concerning readings or missed medication, with appropriate clinical oversight.
- **Advanced Predictive Models:** Develop and validate time-series models for glucose spike forecasting and appropriate cardiovascular risk assessment.

- **Personalized Care Plans:** Enable doctors to provide recommendations based on each patient's health history and monitoring data.

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*Important: Digital Twins Support Doctors—They Do Not Replace Them*

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CuraTwin is intended to support health awareness and clinical decision-making, **not replace professional medical advice, diagnosis, or treatment**. Its predictions and recommendations depend on the quality of the data and the accuracy of the models.
Patients should share relevant symptoms, health changes, medication details, and concerns with their doctor—even when the dashboard indicates a low-risk or normal status. Doctors must interpret the system's outputs alongside clinical examinations, medical history, and appropriate investigations.
**Our vision:** To connect patients and healthcare professionals through a continuously updated digital health profile, supporting more informed, personalized, and proactive diabetes care.